# 2-5 Hypertension v.1

Hypertension is most commonly due to inappropriate depth of anaesthesia or inadequate analgesia.
You should rapidly exclude a problem in adequate oxygen delivery, airway and breathing first.

## START

❶ **Immediate actions**

- Recheck blood pressure AND increase anaesthesia AND reduce stimulus.

❷ **Adequate oxygen delivery**

- Check fresh gas flow for circuit in use AND check measured $F_iO_2$.
- Visual inspection of entire breathing system including valves and connections.
- Rapidly confirm reservoir bag moving OR ventilator bellows moving.

❸ **Airway**

- Check position of airway device and listen for noise (including larynx and stomach).
- Check capnogram shape compatible with patent airway.
- Confirm airway device is patent (consider passing suction catheter).

❹ **Breathing** - exclude **hypoxia** and **hypercarbia** as causes:

- Check chest symmetry, rate, breath sounds, $SpO_2$, measured $VT_{exp}$, $ETCO_2$.
- Feel the airway pressure using reservoir bag and APL valve <3 breaths.

❺ **Circulation**

- Check rate, rhythm, perfusion; increase frequency of BP check.
- Check cuff size and location, consider intra-arterial monitoring.

❻ **Depth**

- Ensure adequate depth of anaesthesia and analgesia.

❼ Consider underlying problem (Box B).

❽ Call for help and consider temporising drug (Box C) if problem not resolving.

### Box A: CRITICAL CHANGES

If problem worsens significantly or a new problem arises, call for **help** and go back to **START** of **1-1 Key Basic Plan.**

### BOX B: POTENTIAL UNDERLYING PROBLEMS

- Inadequate anaesthesia / analgesia (alfentanil can be diagnostic – see Box C for dose)
- Inadequate neuromuscular blockade
- Consider whether you could have made a drug error
- Omission of usual antihypertensives
- Distended bladder
- Vasopressor administered by surgeon
- Surgical tourniquet
- Excess fluid (over-administration / overload / TURP syndrome)
- Medical causes: drug interaction, renal failure, raised intracranial pressure, seizure, thyrotoxicosis, phaeochromocytoma

### BOX C: TEMPORISING DRUGS FOR HYPERTENSION

- Alfentanil 10 $\mu g.kg^{-1}$ (adult 0.5-1 mg)
- Propofol 1 $mg.kg^{-1}$ (adult 50-100 mg)
- Labetolol 0.5 $mg.kg^{-1}$ (adult 25-50mg). Repeat when necessary.
- Esmolol 0.5 $mg.kg^{-1}$ (adult 25-50mg) Follow with infusion.
- Hydralazine 0.1 $mg.kg^{-1}$ (adult 5-10mg)
- Glyceryl trinitrate 0.5-5 $\mu g.kg.min^{-1}$ infusion (adult 2-20 $ml.hr^{-1}$ of 1 $mg.ml^{-1}$ solution)

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